

L VARSHITHA

Bengaluru, Karnataka | +91 9110542033 | Email | LinkedIn | GitHub | LeetCode | Portfolio

Summary

Software Engineer and Computer Science Engineering student with internship experience building backend services, REST APIs, and software components using **Python, Java, and C**. Strong at writing, debugging, and validating software with a solid foundation in **data structures, algorithms, databases, and operating systems**. Experienced in integrating third-party libraries and APIs into larger applications, performing **unit, integration, and API testing**, and identifying edge cases and defects during development. Familiar with **Git, CI/CD, Docker, and Agile** practices, with a focus on building reliable, maintainable, and high-quality software.

Education

Jain (Deemed-to-be) University, Bengaluru

Expected May 2027

B.Tech in Computer Science Engineering (IoT) | CGPA: 9.63/10

– **Relevant Coursework:** Cloud Computing, Machine Learning, Software Engineering, Operating Systems, Computer Networks, DBMS, Object-Oriented Programming, IoT & Embedded Systems.

Technical Skills

Programming Languages: Python, Java, C, TypeScript, JavaScript, SQL

CS Fundamentals: Data Structures Algorithms, OOP, Operating Systems, DBMS, Computer Networks, System Design

Software Development: SDLC, Software Design, Debugging, Issue Investigation, Component Development

Testing Quality: Unit Integration Testing, Functional Testing, API Testing, Test Case Design, Edge Case Boundary Testing, Defect Analysis

Backend APIs: FastAPI, Flask, REST APIs, SQLAlchemy, Pydantic, JWT Authentication, Webhooks

Frontend: React.js, Next.js, Redux

Databases Data: PostgreSQL, MySQL, SQL, Data Validation, Data Pipelines

Cloud: AWS (EC2, S3, Lambda, IAM – Foundational)

Tools Practices: Git, GitHub, Docker, Linux, Postman, PyTest, CI/CD, Agile

Experience

AlgoUniversity

Jun 2026 – Jul 2026

Member of Technical Staff Intern

- Engineered and shipped backend services for **Archie**, an AI-powered System Design mock interview platform, using **Python, FastAPI, SQLAlchemy, PostgreSQL**, and REST APIs to support authentication, session management, chat history, and AI-driven feedback.
- Developed and maintained **PyTest** unit and integration tests for backend services, validating API behavior and preventing regressions while using **ESLint** to enforce frontend code quality before merge.
- Designed and optimized database schemas and REST API endpoints, validating **edge cases, boundary conditions, authentication flows, and error scenarios** to improve application reliability and backend data consistency.
- Collaborated with frontend engineers through development, debugging, testing, and code reviews to deliver maintainable, production-ready features in an Agile environment.

Lumbini Technologies

Feb 2026 – Apr 2026

Software Engineer Intern

- Built a full-stack Event Management System using **Next.js, TypeScript, Prisma, and PostgreSQL**, implementing role-based access control and backend workflows for multiple user types.
- Designed and validated RESTful APIs for **event management, authentication, file uploads, and dashboards** using Postman, verifying request/response behavior, error handling, and integration flows.
- Integrated the **OpenAI API** with LaTeX to automate event proposal and report generation, improving workflow efficiency while handling API responses and document-generation edge cases.
- Optimized database queries and implemented **JWT-based authentication**, debugging integration issues and delivering features through code reviews, sprint planning, and an **Agile SDLC** workflow.

Projects

Industrial Digital Twin & Automated Sensor Validation Framework

Python, FastAPI, SSE, REST APIs | 2025

- Engineered a real-time digital twin that ingests live multi-sensor data streams and runs an automated test-and-validation framework to flag anomalous sensor readings against expected behavior.
- Built low-latency monitoring pipelines using Server-Sent Events (SSE) and REST APIs for continuous state verification, regression checks, and system health monitoring.
- Designed automated decision logic to identify edge cases and filter false positives, improving test signal reliability without heavy infrastructure overhead.

AI-Based Task Manager

Python, Flask, React.js, spaCy | Jun 2025 – Aug 2025

- Built a full-stack application that converts natural-language input into organized, prioritized tasks, streamlining daily task management workflows for end users.
- Developed an interactive React.js frontend with reusable components and real-time filtering, backed by Flask REST APIs and spaCy NLP pipelines that extract intent, entities, priorities, and deadlines from free-text input.

Smart Garbage Classifier

Python, PyTorch, Flask | Mar 2025 – May 2025

- Built a deep-learning computer vision system with a Flask REST API backend to automate waste segregation, exposing model-inference endpoints for web clients.
- Trained, optimized, and rigorously evaluated a PyTorch CNN with OpenCV preprocessing, achieving **93.4% validation accuracy**, and packaged the model for production deployment.

Leadership & Achievements

- **Technical Lead, Zigbee Club, Jain University (2024–Present):** led technical workshops and mentored 100+ students in IoT, Embedded Systems, and Wireless Communication.
- **Spectre Hackathon (2026):** Top 20 teams, national-level hackathon.
- **RPC Edulateral Hackathon (2024):** Top 12 finalist for technical problem solving.
- **Pitch & Sustain Competition (2025):** Awarded First Prize for an innovative solution.